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The Edge Awakens: Why Power Semiconductors Are the Next Scarcity Trade

Jordi Visser

JVisser@22VResearch.com

22vresearch.com

Artificial intelligence had been dominated by one story: compute. GPUs, memory, and massive model training captured nearly all the attention in performance through the end of 2025. But as I outlined in my recent thematic work, *"The Whole Rack"*, which expanded the AI opportunity from GPUs to the full infrastructure stack, and *"Edge AI Investment Universe"*, which mapped the system-level bottlenecks across compute, power, and deployment, the real shift underway is broader driven by the needs of the agentic world and the AI physical upgrade. This week's signals from Intel, Texas Instruments, and NVIDIA's reported 800V DC power push in South Korea reinforce that view. The next phase of AI will be defined by where intelligence happens and how efficiently it is powered. Edge devices and power efficiency are now converging into one investment theme, and power semiconductors sit directly in the middle.

Intel made the edge thesis explicit in earnings this week. As AI moves from foundational model training to inference and agentic workloads, more computation needs to happen closer to the end user. Intel said the "next wave of AI will bring intelligence closer to the end user," increasing demand for CPUs, wafers, and advanced packaging. It also emphasized that the CPU is becoming central to edge AI computing as orchestration tasks and agentic workloads scale. That matters because AI inference is continuous, distributed, and latency-sensitive. It needs CPUs, connectivity, packaging, and efficient power management across far more locations than the original GPU training cycle.

Texas Instruments adds the physical-world evidence from its earnings. TI's industrial demand is broadening, with management pointing to several quarters of improvement and strength tied to industrial customers and data centers. That industrial strength increasingly reflects smarter machines, buildings, factories, vehicles, and grid infrastructure. TI has also described edge AI as already present in devices, enabled by microcontrollers with AI accelerators and neural processing units. These systems bring intelligence into equipment that must run efficiently, reliably, and often with tight power constraints.

Put Intel and TI together and the message becomes clear: edge AI is starting to move from concept to deployment. Intel is seeing the compute side through CPUs and agentic workloads. TI is seeing the device side through industrial electronics, embedded systems, analog chips, and power management. The common thread is that intelligence is spreading outward from the cloud into the physical world.

That leads directly to the power problem. Recent South Korean media reports say NVIDIA has approached Korean power equipment companies about designing infrastructure around 800V DC architectures for next-generation AI data centers. The goal is to reduce conversion losses, cut copper intensity, simplify power distribution, and support far higher rack power densities. This is an important signal. NVIDIA is treating power delivery as a core scaling challenge for AI infrastructure.

The move from traditional lower-voltage systems toward 800V DC changes the power stack. Higher-voltage architectures require more sophisticated power devices, better conversion, advanced protection, and tighter system design. This increases the importance of silicon carbide, gallium nitride, high-performance analog, power modules, solid-state transformers, and power management ICs. In other words, the AI infrastructure trade is broadening from GPUs and memory into the components that move, convert, regulate, and protect electricity.

This is why power semiconductors are becoming a central AI beneficiary. Every step of the AI buildout requires more efficient power conversion. Data centers need it to handle megawatt-scale racks. Edge devices need it to run intelligence locally without wasting energy. Industrial systems need it for motors, sensors, robotics, automation, and grid equipment. Vehicles need it as cars become rolling compute platforms. The more AI spreads, the more power management becomes mission-critical.

The most important part of this thesis is that the edge and the data center are connected. The cloud trains and coordinates large models, while the edge executes more intelligence closer to users, machines, and devices. Both sides increase demand for power semiconductors. Centralized AI needs higher-voltage, higher-density power systems. Distributed AI needs efficient local power management across billions of endpoints. The same macro trend supports both.

This also changes the way investors should think about cyclicalities. Power semiconductors have historically been treated as industrial and auto cycle stocks, tied to the familiar four-year rhythm of inventory corrections, factory slowdowns, and auto production cycles. That perception still exists after the 2024–2025 industrial slowdown. At the same time, the structural tailwind from AI is beginning to reshape demand. The end markets are expanding beyond autos and factories into AI data centers, edge devices, robotics, and grid infrastructure. The cycle remains, but the underlying demand base is part of a cycle the world has never experienced and becoming broader and more durable.

The geopolitical overlay makes the argument even stronger. Sovereign AI is becoming a national priority, with countries building local compute capacity, localized data centers, and more resilient technology supply chains. Those systems require localized electricity infrastructure, grid upgrades, and advanced power conversion. Power semiconductors are therefore part of the “China + 1” strategy and increasingly tied to national infrastructure planning. As countries invest in energy systems to support AI, the power stack becomes a strategic layer of the technology ecosystem.

This creates a powerful demand loop. More AI adoption creates more inference. More inference creates more edge compute. More edge compute creates more power demand. More power demand creates greater need for efficiency. Greater efficiency requires more advanced power semiconductors. This loop is still early because most of the world’s machines, vehicles, buildings, factories, and devices have not yet been rebuilt around AI. This is the base of Jensen Huang’s Five Layer Cake analogy

That is the key investment point. The first AI wave rewarded the companies closest to model training. The next wave should broaden toward the companies that enable AI to scale in the real world. That includes CPUs, MCUs, analog, power management, advanced packaging, optics, cooling, electrical equipment, and grid infrastructure. Power semiconductors are especially important because they sit at the intersection of compute growth, energy demand, edge deployment, and national infrastructure investment.

Intel’s earnings commentary shows that the compute architecture is shifting. Texas Instruments shows that industrial edge demand is broadening. NVIDIA’s reported 800V DC push shows that power architecture itself is being redesigned. These are three different signals pointing in the same direction: AI is becoming more distributed, more physical, and more power-intensive.

The market is still early in recognizing this. Investors understand GPUs and memory. They are beginning to understand optics and cooling. The next layer is power. As AI moves from centralized training to persistent inference and then into agentic systems, power efficiency becomes one of the most valuable bottlenecks in the entire stack.

The conclusion is simple: the next phase of AI is about powering intelligence everywhere. Power semiconductors are evolving into core infrastructure for the AI age. And if edge AI is only beginning, then the power semiconductor cycle is likely only beginning too.

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